

Student:	Email:
Mark Jordan	jordanm14@nku.edu

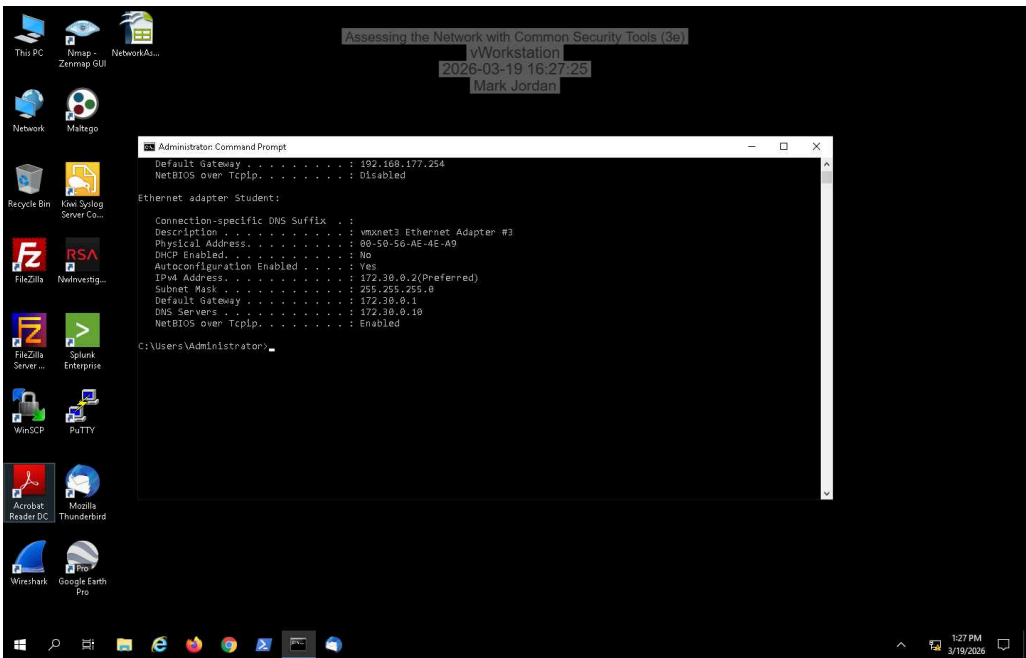
Time on Task:	Progress:
2 hours, 57 minutes	100%

Report Generated: Saturday, March 21, 2026 at 9:17 AM

Section 1: Hands-On Demonstration

Part 1: Explore the Local Area Network

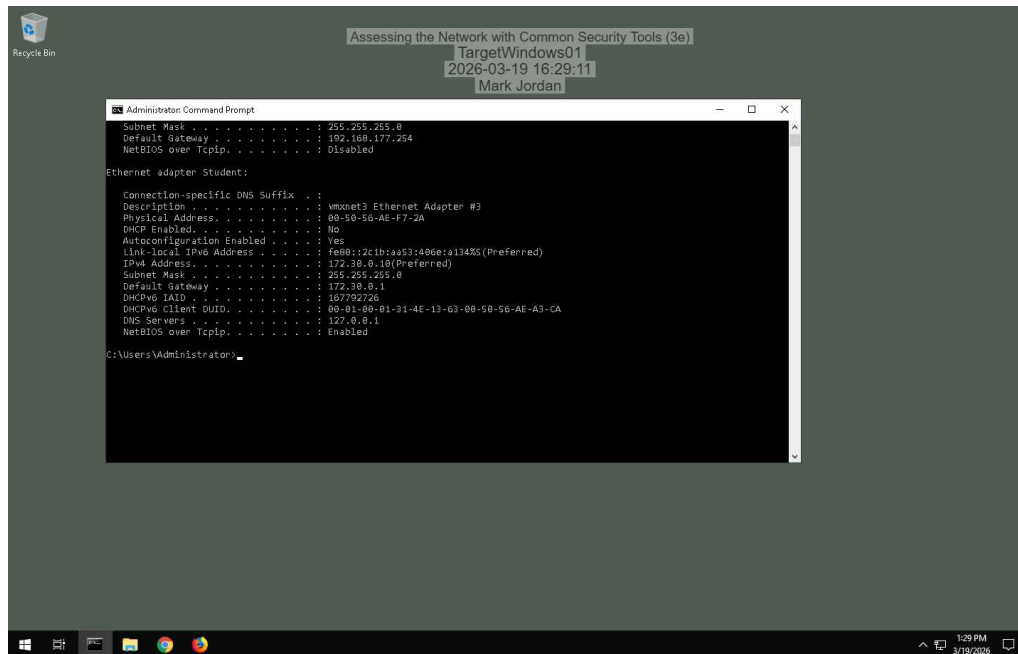
4. Make a screen capture showing the ipconfig results for the Student adapter on the vWorkstation.



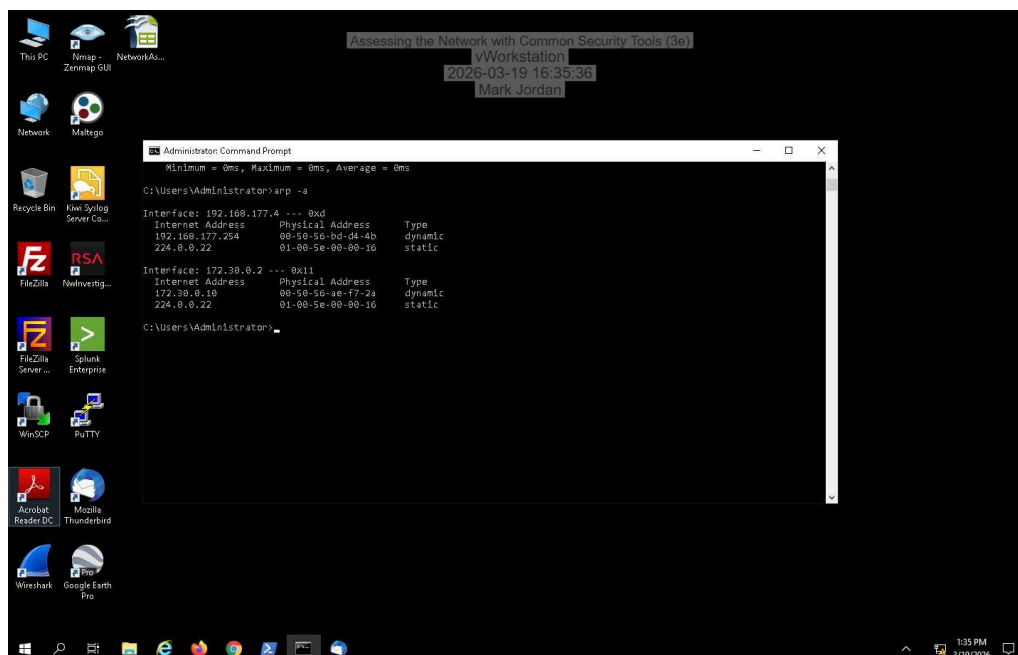
Assessing the Network with Common Security Tools (3e)

Network Security, Firewalls, and VPNs, Third Edition - Lab 01

7. Make a screen capture showing the **ipconfig** results for the Student adapter on **TargetWindows01**.



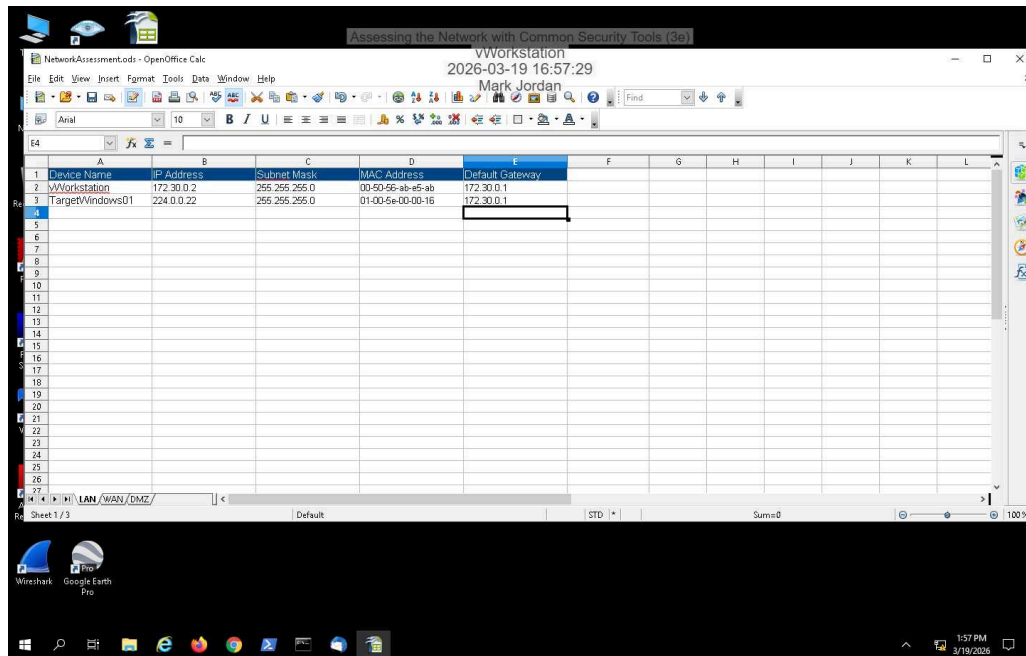
15. Make a screen capture showing the updated ARP cache on the vWorkstation.



Assessing the Network with Common Security Tools (3e)

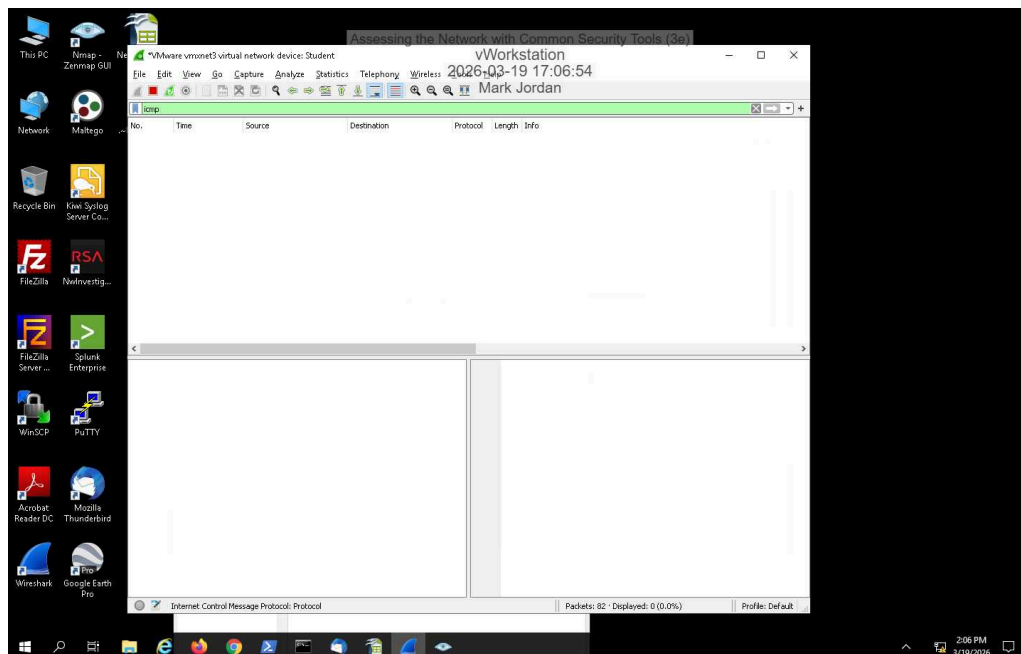
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19. Make a screen capture showing the **completed LAN tab** of the Network Assessment spreadsheet.

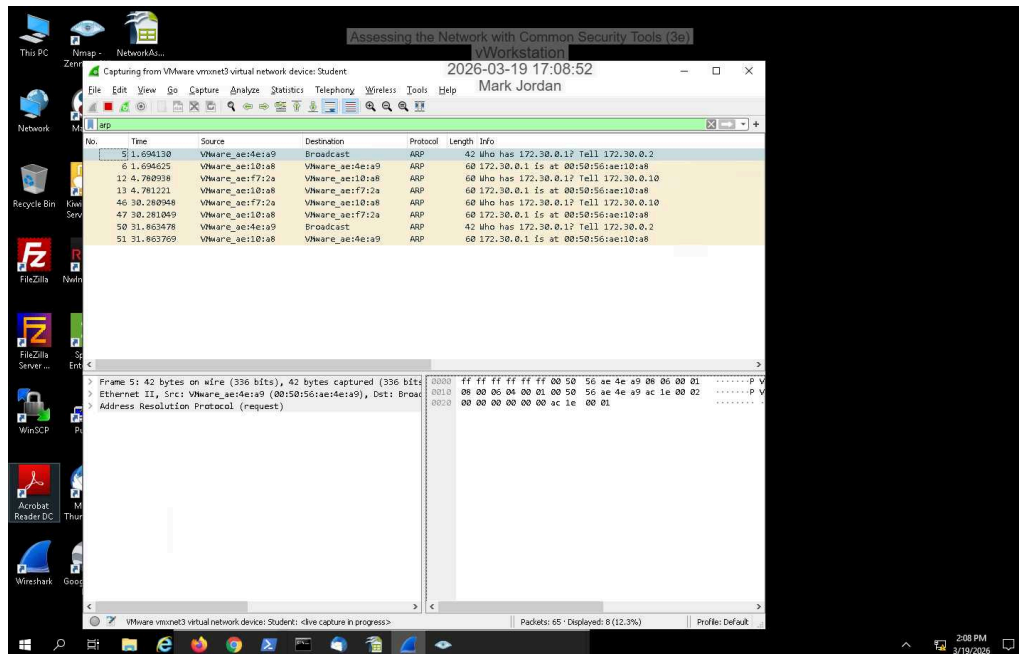


Part 2: Analyze Network Traffic

9. Make a screen capture showing the **ICMP filtered results** in Wireshark.



12. Make a screen capture showing the ARP filtered results in Wireshark.



18. Compare the Regular scan results for ICMP and ARP traffic with the results from the Ping scan.

With a regular scan, I was able to identify one ICMP packet, however it looks like it was an error packet "Destination unreachable". I was also able to see a lot of ARP traffic. Whereas in the ping scan, there were no ICMP packets and about the same amount of ARP traffic. ARP filters appear to be more comprehensive and provide more detail on MAC addresses

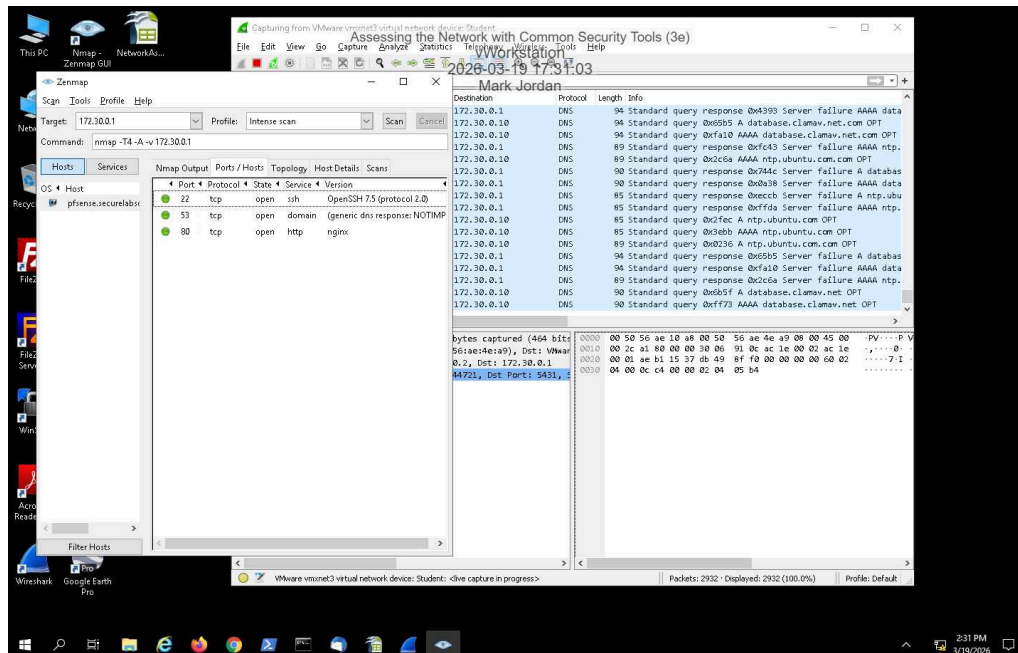
24. Compare the Intense scan results with the results from the Ping scan.

The intense scan provides a more in depth layout of the network. It provides IP service versions, open ports operating systems and provides a network topology to identify vulnerabilities. Based on the Wireshark activity, it is a more noisy process. The ping scan only provided minimal information such as a MAC address and latency times.

Assessing the Network with Common Security Tools (3e)

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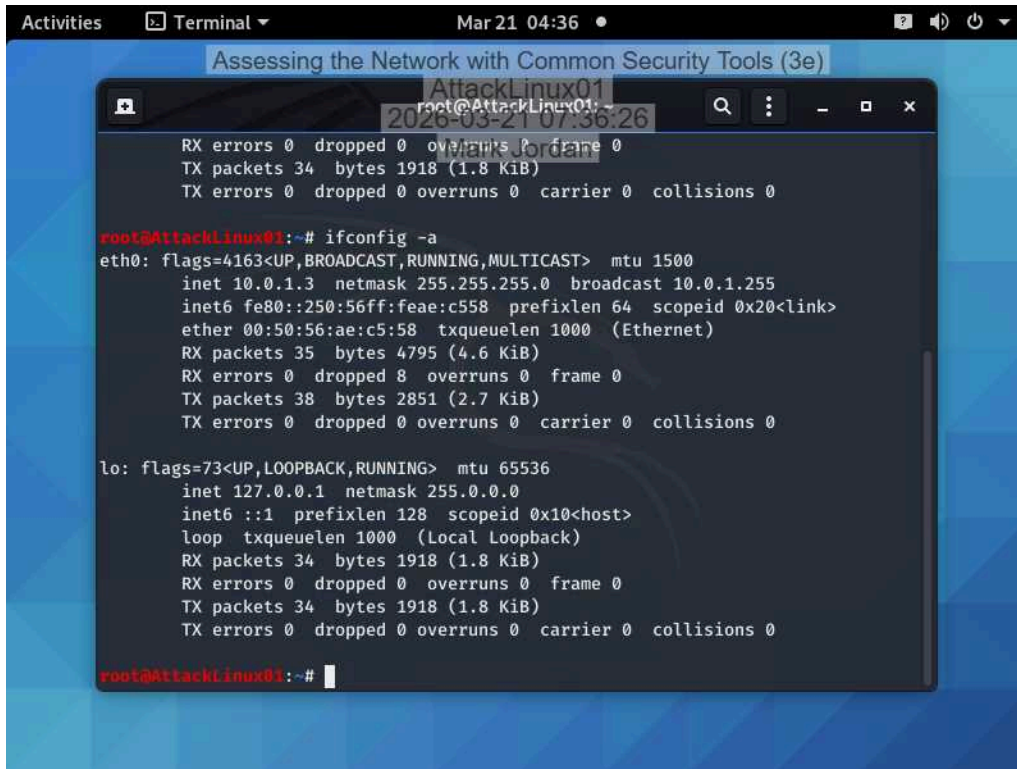
28. Make a screen capture showing the contents of the Ports/Hosts tab.



Section 2: Applied Learning

Part 1: Explore the Wide Area Network

6. Make a screen capture showing the `ifconfig` results on `AttackLinux01`.



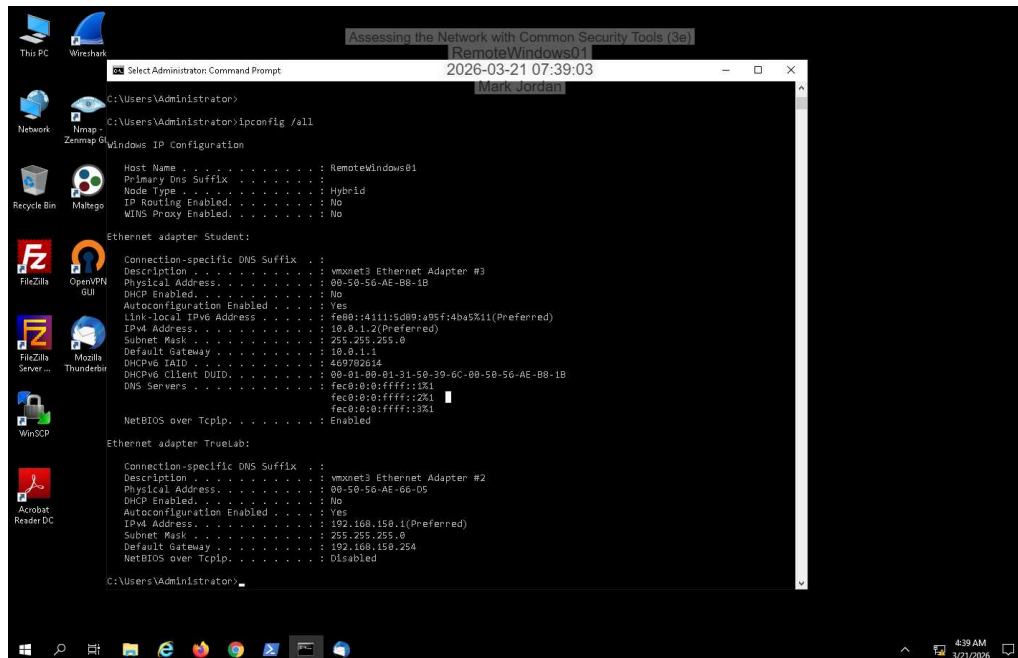
The screenshot shows a terminal window titled "Terminal" with the date and time "Mar 21 04:36". The terminal is running on a system named "AttackLinux01". The user is at the root prompt and has executed the command `ifconfig -a`. The output shows the configuration for the `eth0` and `lo` interfaces. The `eth0` interface is up and running, with an IP address of `10.0.1.3` and a netmask of `255.255.255.0`. The `lo` interface is also up and running, with an IP address of `127.0.0.1` and a netmask of `255.0.0.0`.

```
Activities Terminal Mar 21 04:36
Assessing the Network with Common Security Tools (3e)
AttackLinux01
root@AttackLinux01:~# ifconfig -a
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.1.3 netmask 255.255.255.0 broadcast 10.0.1.255
    inet6 fe80::250:56ff:feae:c558 prefixlen 64 scopeid 0x20<link>
    ether 00:50:56:ae:c5:58 txqueuelen 1000 (Ethernet)
    RX packets 35 bytes 4795 (4.6 KiB)
    RX errors 0 dropped 8 overruns 0 frame 0
    TX packets 38 bytes 2851 (2.7 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

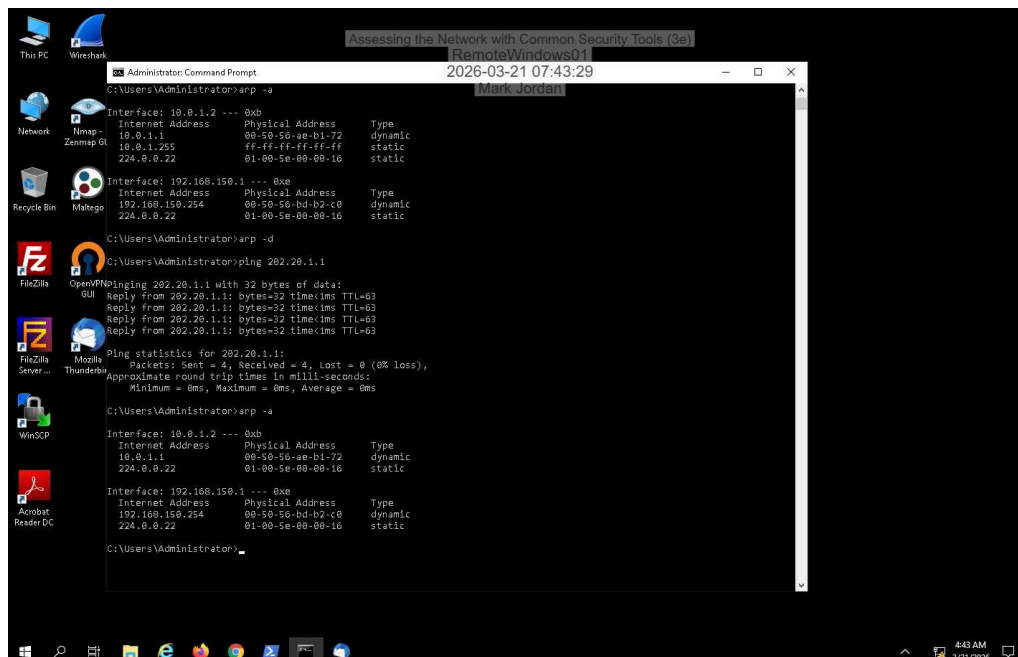
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 34 bytes 1918 (1.8 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 34 bytes 1918 (1.8 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@AttackLinux01:~#
```

12. Make a screen capture showing the ipconfig results on RemoteWindows01.



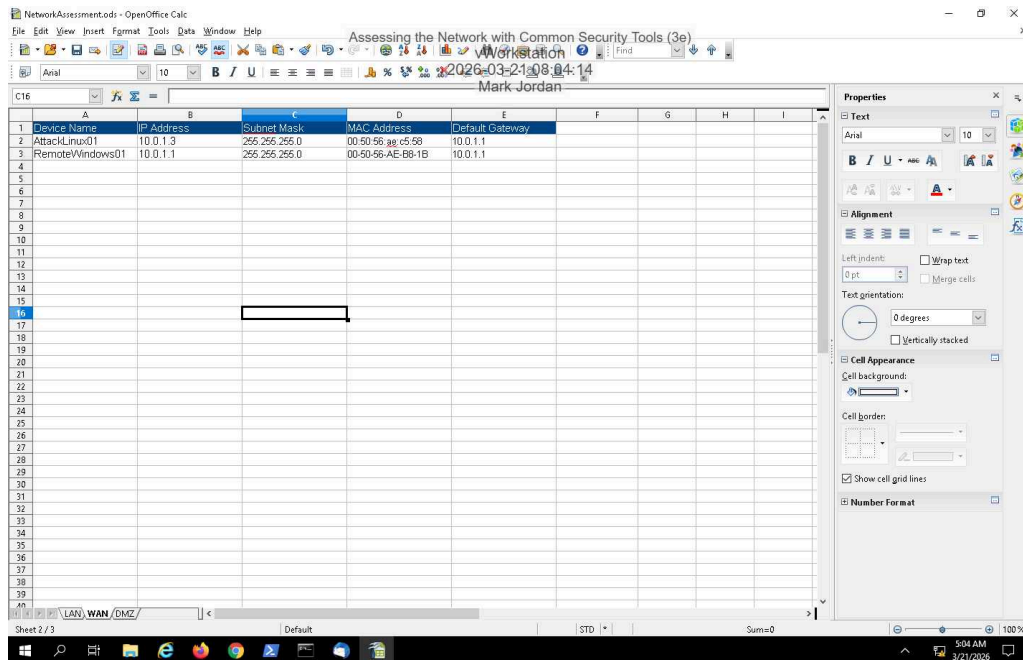
18. Make a screen capture showing the updated ARP cache on RemoteWindows01.



Assessing the Network with Common Security Tools (3e)

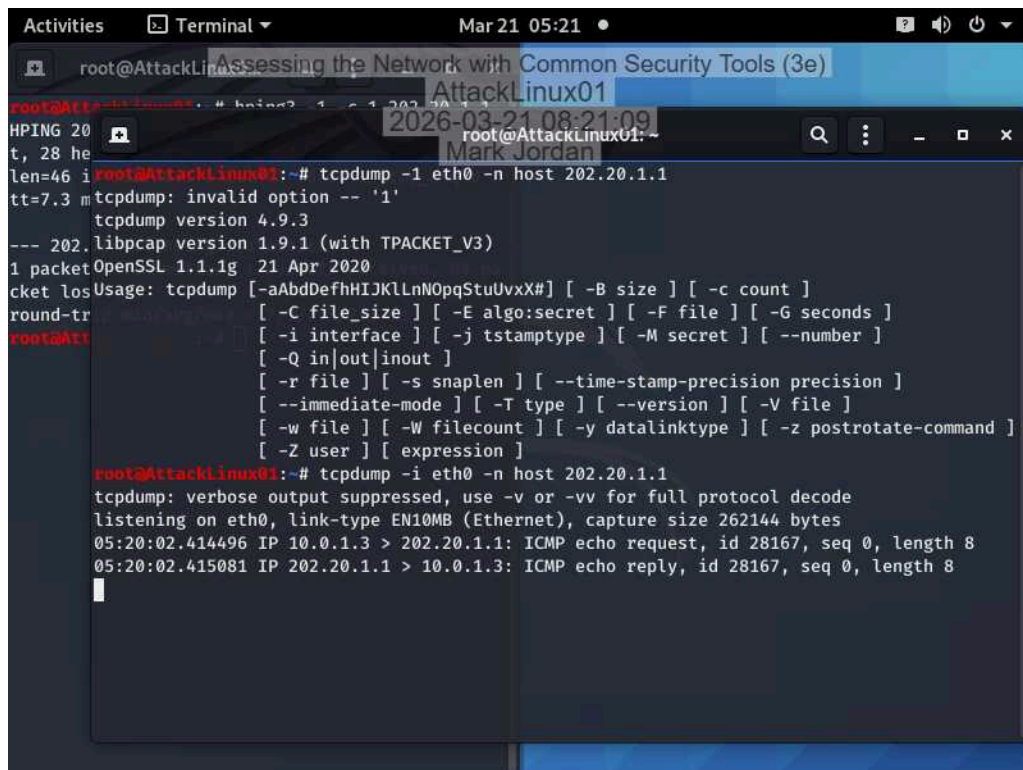
Network Security, Firewalls, and VPNs, Third Edition - Lab 01

22. Make a screen capture showing the **completed WAN tab** of the Network Assessment spreadsheet.

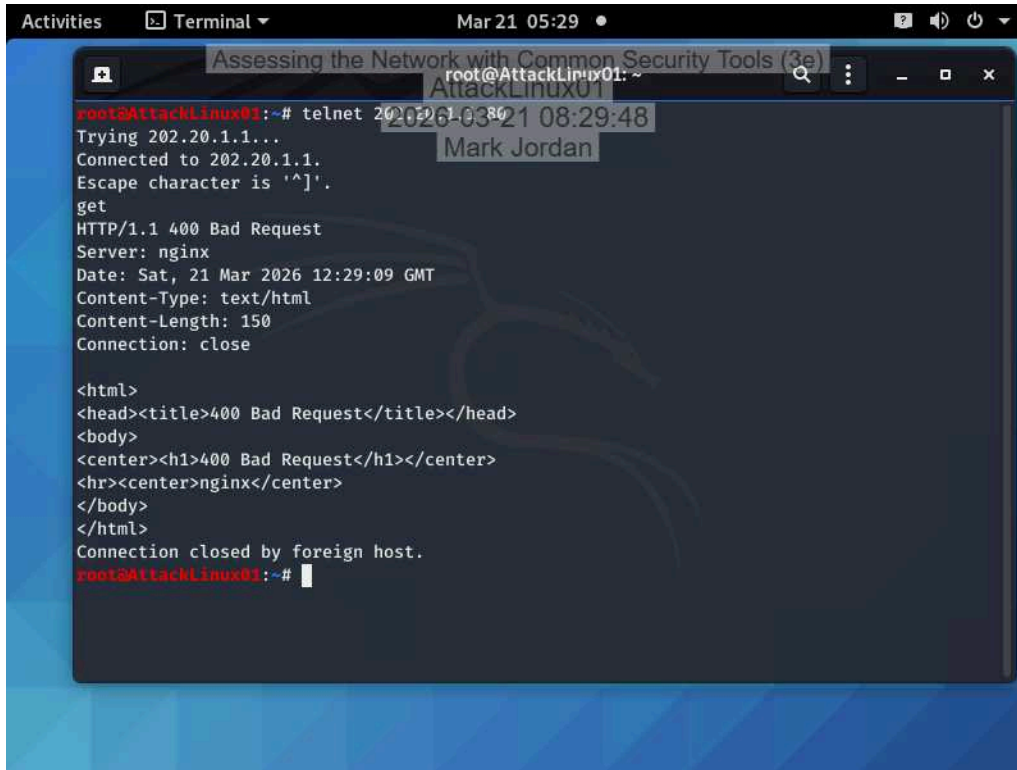


Part 2: Analyze Network Traffic

9. Make a screen capture showing **tcpdump** echo back the captured packets.



12. Make a screen capture showing the attempted three-way handshake in tcpdump.
17. Make a screen capture showing the results of the get command.



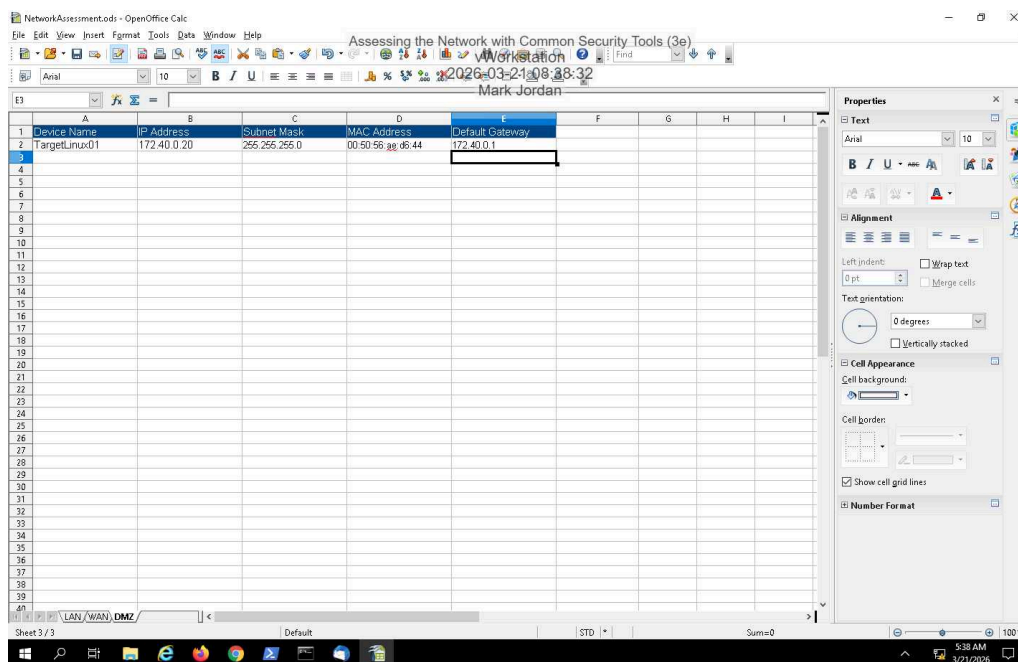
The screenshot shows a terminal window titled "Terminal" with the date and time "Mar 21 05:29". The terminal output is as follows:

```
root@AttackLinux01: ~  
root@AttackLinux01:~# telnet 202.20.1.1 80  
Trying 202.20.1.1...  
Connected to 202.20.1.1.  
Escape character is '^]'.  
get  
HTTP/1.1 400 Bad Request  
Server: nginx  
Date: Sat, 21 Mar 2026 12:29:09 GMT  
Content-Type: text/html  
Content-Length: 150  
Connection: close  
  
<html>  
<head><title>400 Bad Request</title></head>  
<body>  
<center><h1>400 Bad Request</h1></center>  
<hr><center>nginx</center>  
</body>  
</html>  
Connection closed by foreign host.  
root@AttackLinux01:~#
```

Section 3: Challenge and Analysis

Part 1: Explore the DMZ

Make a screen capture showing the **completed DMZ tab** of the **NetworkAssessment** spreadsheet.



Part 2: Perform Reconnaissance on the Firewall

Briefly summarize and analyze your findings in a technical memo to your boss.

To: Boss
From: Mark Jordan
Date: 03/21/2026

Today I was tasked with scanning the pfSense firewall in order to identify any packets sent to the firewall and identify any open ports on the pfSense firewall. When Utilizing the regular scan on Zenmap and Wireshark tool, I was able to identify multiple packets being sent to the pfSense firewall (IP 202.20.1.1). I was able to identify 13 ICMP packets, 2 ARP packets and 12 DNS Packets. There are a total of 3 ports open. Ports 22, 53, and 80.

Thank you,
Mark Jordan